Potential and Vector Potential with Multipole Expansion

Yen-Ting Wu (Lecture Note)

Potential and Vector Potential

All the equations we will discuss in this note:

$$\begin{cases} V(\mathbf{r}) = \dfrac{1}{4\pi\epsilon_0}\displaystyle\int \dfrac{\varrho(\mathbf{r}')}{\boldsymbol{\mathscr{r}}}\, d\tau' \\ V_{\text{dip}}(\mathbf{r}) = \dfrac{1}{4\pi\epsilon_0}\displaystyle\int \dfrac{\mathbf{P}(\mathbf{r}')\cdot\hat{\boldsymbol{\mathscr{r}}}}{\boldsymbol{\mathscr{r}}^2}\, d\tau' \end{cases} \qquad \begin{cases} \mathbf{A}(\mathbf{r}) = \dfrac{\mu_0}{4\pi}\displaystyle\int \dfrac{\mathbf{J}(\mathbf{r}')}{\boldsymbol{\mathscr{r}}}\, d\tau' \\ \mathbf{A}_{\text{dip}}(\mathbf{r}) = \dfrac{\mu_0}{4\pi}\displaystyle\int \dfrac{\mathbf{M}(\mathbf{r}')\times\hat{\boldsymbol{\mathscr{r}}}}{\boldsymbol{\mathscr{r}}^2}\, d\tau' \end{cases}$$

Note that the cases here we just focus on the static situations, or the varied field may influence another one.

1 Electric Potential and Dipole Contribution

1.1 Electric Potential

By Coulomb's law, the electric field is defined as:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{\varrho(\mathbf{r}')\hat{\boldsymbol{\mathscr{r}}}}{\boldsymbol{\mathscr{r}}^2}\, d\tau'. \tag{1}$$

Then Gauss's law gives that

$$\boldsymbol{\nabla}\cdot\mathbf{E}(\mathbf{r}) = \frac{\varrho(\mathbf{r})}{\epsilon_0}. \tag{2}$$

Because the electric field is conservative in electrostatics, that is

$$\oint_{\mathcal{P}} \mathbf{E}(\mathbf{r})\cdot d\boldsymbol{\ell} = 0, \quad \boldsymbol{\nabla}\times\mathbf{E} = \mathbf{0},$$

so we can find a potential function(ϕ, or V) of the electric field which is defined as

$$V(\mathbf{r}) = -\int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E}(\mathbf{r}')\cdot d\boldsymbol{\ell}', \quad \Leftrightarrow \quad \mathbf{E} = -\boldsymbol{\nabla}V$$

in Griffith's textbook. But in general, I'm going to use the Green function to solve the partial differential equation. We substitute $\mathbf{E} = -\boldsymbol{\nabla} V$ into the Gauss's law:

$$\nabla^2 V(\mathbf{r}) = -\frac{\varrho(\mathbf{r})}{\epsilon_0}, \tag{3}$$

and this is called "**Poisson's Equation**". By the Delta Function identity (iv), we've known that

$$\nabla^2\left(\frac{1}{\mathscr{r}}\right) = -4\pi\delta^3(\boldsymbol{\mathscr{r}}), \tag{4}$$

and the Green function $G(\mathbf{r}, \mathbf{r}')$ is defined as

$$\begin{cases} \mathcal{L}G(x, x') = -\delta(x - x') \\ \\ \mathcal{L}G(\mathbf{r}, \mathbf{r}') = -\delta^3(\boldsymbol{\mathscr{r}}) \end{cases}, \quad \mathcal{L} \text{ is a linear operator(differential)} \tag{5}$$

This implies that when we want to solve an equation for $\phi(x)$, which satisfies

$$\mathcal{L}\phi(x) = f(x), \tag{6}$$

if we can find a $G(x, x') = -\delta(x - x')$, then we can further write down the solution to the equation that

$$\begin{aligned} \phi(x) &= -\int G(x, x')f(x')dx' \\ \Rightarrow \mathcal{L}\phi(x) &= -\int \mathcal{L}G(x, x')f(x')dx' \\ &= -\int \big[-\delta(x - x')\big]f(x')dx' = f(x) \end{aligned}$$

Back to (4), it can be rewritten

$$\nabla^2\left(\frac{1}{4\pi\mathscr{r}}\right) = -\delta^3(\boldsymbol{\mathscr{r}}), \tag{7}$$

therefore, we can determine the Green function in (7):

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi\mathscr{r}}. \tag{8}$$

Then we can obtain the solution to (3), and we substitute (8) into it, it reads

$$V(\mathbf{r}) = -\int G(\mathbf{r}, \mathbf{r}')\left[-\frac{\varrho(\mathbf{r}')}{\epsilon_0}\right]d\tau' = \int \left(\frac{1}{4\pi\mathscr{r}}\right)\left[\frac{\varrho(\mathbf{r}')}{\epsilon_0}\right]d\tau'. \tag{9}$$

This yields the final result:

$$\boxed{V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{\varrho(\mathbf{r}')}{\mathscr{r}}d\tau'} \tag{10}$$

1.2 Multipole Expansion

We've known the general solution to Poisson's Equation $V(\mathbf{r})$, and we can further expand the term of it as we've defined $\boldsymbol{\mathscr{r}} = \mathbf{r} - \mathbf{r}'$. So,

$$|\boldsymbol{\mathscr{r}}|^2 = \mathscr{r}^2 = r^2 + \left(r'\right)^2 - 2rr'\cos\alpha = r^2\left[1 + \left(\frac{r'}{r}\right)^2 - 2\left(\frac{r'}{r}\right)\cos\alpha\right],$$

where α is the angle between the vectors $\mathbf{r}$ and $\mathbf{r}'$. We assume that $r' < r$, which means that we observe the source in the observation coordinates far from the source. This yields

$$\frac{1}{\mathscr{r}} = \frac{1}{r}\left[1 + \left(\frac{r'}{r}\right)^2 - 2\left(\frac{r'}{r}\right)\cos\alpha\right]^{-1/2} = \frac{1}{r}(1+\epsilon)^{-1/2}, \tag{11}$$

where

$$\epsilon := \left(\frac{r'}{r}\right)\left(\frac{r'}{r} - 2\cos\alpha\right). \tag{12}$$

We've known that[1]

$$(1+\epsilon)^{-1/2} = \sum_n (-1)^n \frac{(2n)!}{(n!)^2 \times 4^n} x^n = 1 - \frac{1}{2}\epsilon + \frac{3}{8}\epsilon^2 - \frac{5}{16}\epsilon^3 + \cdots,$$

and we substitute (12) into this expansion, then you can choose to "TRUST" the result in Griffith's textbook that

$$\boxed{\frac{1}{\mathscr{r}} = \frac{1}{r}\sum_{n=0}^{\infty}\left(\frac{r'}{r}\right)^n P_n(\cos\alpha).} \tag{13}$$

$P_n(\cos\alpha)$ is the Legendre polynomial[2]. Or, you can look up my personal notes for *Spherical Harmonic Function* and choose to "BELIEVE" in me:) That'll be great! In general, the expansion of $1/\mathscr{r}$ is:

$$\frac{1}{\mathscr{r}} = 4\pi \sum_{n=0}^{\infty}\sum_{m=-n}^{n} \frac{1}{2n+1}\frac{(r')^n}{r^{n+1}} Y_{nm}(\Omega)Y^*_{nm}(\Omega'), \quad r' < r,$$

where $\Omega = (\alpha, \phi)$ and $\Omega' = (\alpha', \phi')$. When we let the source $\mathbf{r}'$ be along z-axis, then $\Omega' = (0, \phi')$ and $m = 0$. We obtain

$$\begin{cases} Y_{n0}(\Omega) = \sqrt{\dfrac{2n+1}{4\pi}} P_n(\cos\alpha) \\ Y^*_{n0}(0, \phi') = \sqrt{\dfrac{2n+1}{4\pi}}. \end{cases} \tag{14}$$

[1]See *Mathematical Methods for Taylor's Mechanics, Yen-Ting Wu* Eq.(1.182).
[2]See *Separation of Variables and Legendre Polynomial, Yen-Ting Wu.*

Therefore,

$$\begin{aligned}
\frac{1}{\mathscr{r}} &= 4\pi \sum_{n=0}^{\infty} \sum_{m=-n}^{n} \frac{1}{2n+1} \frac{(r')^n}{r^{n+1}} Y_{nm}(\Omega) Y^*_{nm}(\Omega'), \qquad m=0 \\
&= \sum_{n=0}^{\infty} \frac{4\pi}{2n+1} \frac{(r')^n}{r^{n+1}} \left(\sqrt{\frac{2n+1}{4\pi}} P_n(\cos\alpha)\right) \left(\sqrt{\frac{2n+1}{4\pi}}\right) \\
&= \sum_{n=0}^{\infty} \frac{(r')^n}{r^{n+1}} P_n(\cos\alpha),
\end{aligned}$$

as before. But this is not the point in this note. We substitute (13) into the general solution to the potential $V(\mathbf{r})$, then

$$\boxed{V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \int (r')^n P_n(\cos\alpha) \varrho(\mathbf{r}')\, d\tau'.} \tag{15}$$

1.3 Dipole Contribution to the Potential

We expand the result (15) in terms of $\cos\alpha$, we obtain:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{1}{r} \int \varrho(\mathbf{r}')\, d\tau' + \frac{1}{r^2} \int r' \cos\alpha\, \varrho(\mathbf{r}')\, d\tau' + \text{Quadrupole} + \cdots \right] \tag{16}$$

The first term is called Monopole potential $V_{\text{mon}}(\mathbf{r})$ and the second one is called Dipole potential $V_{\text{dip}}(\mathbf{r})$. They are respectively:

$$\begin{cases} V_{\text{mon}}(\mathbf{r}) = \dfrac{1}{4\pi\epsilon_0} \dfrac{1}{r} \displaystyle\int \varrho(\mathbf{r}')\, d\tau' \\ V_{\text{dip}}(\mathbf{r}) = \dfrac{1}{4\pi\epsilon_0} \dfrac{1}{r^2} \displaystyle\int r' \cos\alpha\, \varrho(\mathbf{r}')\, d\tau'. \end{cases} \tag{17}$$

The monopole means that there is just a single point charge that contributes to the electric field; the dipole means that there are two charges with different electricity. This causes the polarization, and the charges are ”polarized”.
We can rewrite the potential of the dipole as the form:

$$V_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{1}{r^2} \int \underbrace{r' \cos\alpha}_{\mathbf{r}'\cdot\hat{\mathbf{r}}}\, \varrho(\mathbf{r}')\, d\tau' = \frac{1}{4\pi\epsilon_0} \frac{\hat{\mathbf{r}}}{r^2} \cdot \int \mathbf{r}'\, \varrho(\mathbf{r}')\, d\tau', \tag{18}$$

where we define

$$\mathbf{p} \coloneqq \int \mathbf{r}'\, \varrho(\mathbf{r}')\, d\tau', \tag{19}$$

which is called the **Dipole moment**. Therefore, (18) is rewritten in the form

$$V_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\frac{\mathbf{p}\cdot\hat{\mathbf{r}}}{r^2}. \tag{20}$$

Note that the expansion of $1/\boldsymbol{\mathscr{r}}$ is developed with the condition that we let the source $\mathbf{r}'$ be long the specific axis of the coordinates. Therefore, in general, we can surely write the potential of an individual dipole located at an arbitrary position in space and oriented in an arbitrary direction in an orthogonal coordinate system, rather than restricting the dipole to lie along the z-axis:

$$V_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\frac{\mathbf{p}\cdot\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}. \tag{21}$$

This equation shows the property of an individual dipole, in other words, we can further discuss a continuous medium or region that the dipoles distribute to it, so the dipole moment of a infinitesimal volume is $\mathbf{p} = \mathbf{P}d\tau$, where $\mathbf{P}(\mathbf{r})$ is the dipole moment per unit volume. So, the dipole potential can be written as

$$\boxed{V_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int\frac{\mathbf{P}(\mathbf{r}')\cdot\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}\,d\tau'} \tag{22}$$

2 Magnetic Vector Potential and Dipole Contribution

2.1 Magnetic Vector Potential

By Biot-Savart's law, the magnetic field is:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi}\int\frac{\mathbf{J}(\mathbf{r}')\times\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}d\tau \tag{23}$$

The divergence & curl of the field is zero[3]:

$$\begin{cases}\boldsymbol{\nabla}\cdot\mathbf{B}(\mathbf{r}) = 0\\ \boldsymbol{\nabla}\times\mathbf{B}(\mathbf{r}) = \mu_0\mathbf{J}.\end{cases} \tag{24}$$

In comparison with the curl of the electric field $\boldsymbol{\nabla}\times\mathbf{E} = 0$, there's a corresponded potential $V(\mathbf{r})$ to the specific electric field. So, the divergence of the magnetic field gives the inspiration that "Can we find some potential that can describe the contribution to

[3]See *Derivation of Maxwell's Equations, Yen-Ting Wu.*

the field and be symmetric to the electric field?" That's going to be the **vector potential** $\mathbf{A}(\mathbf{r})$ which satisfies:

$$\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A} \Leftrightarrow \boldsymbol{\nabla} \cdot \mathbf{B} = \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \mathbf{A}) = 0. \tag{25}$$

Then the curl of the field in the form of the vector potential is given by:

$$\boldsymbol{\nabla} \times \mathbf{B} = \boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \mathbf{A}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J}, \tag{26}$$

The first term $\boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \mathbf{A})$ of the expression of the curl of curl $\mathbf{A}$ can be eliminated. Note that the vector potential is not unique, like the potential of a conservative field $V(\mathbf{r}) \to V_0 + C$. Therefore, the vector potential can be written as

$$\mathbf{A} = \mathbf{A}_0 + \boldsymbol{\nabla}\lambda,$$

where λ is a "Good" continuous function. Then the divergence of it is:

$$\boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot \mathbf{A}_0 + \boldsymbol{\nabla}\lambda. \tag{27}$$

Without loss of the generality, the key point is the *Gauge* we choose (Coulomb gauge), which means that although something is not unique (like the energy flux $\mathbf{S}$ is also not unique), we can always find some function or vector to make something easier to deal with. This fact gives that we can let the vector potential $\mathbf{A}$ be divergenceless when

$$\boldsymbol{\nabla}\lambda = -\boldsymbol{\nabla} \cdot \mathbf{A}_0 \Leftrightarrow \boldsymbol{\nabla} \cdot \mathbf{A} = 0. \tag{28}$$

No that we don't "need" to do this, and the vector potential $\mathbf{A}$ may not be this form; however, we can "choose" to do that, at least I don't want to look for trouble to me. Hence, (26) becomes

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}. \tag{29}$$

This is the Poisson equation of the vector potential. By (8), the solution to $\mathbf{A}$ is

$$\mathbf{A}(\mathbf{r}) = -\int G(\mathbf{r}, \mathbf{r}')\big[-\mu_0 \mathbf{J}(\mathbf{r}')\big]\, d\tau' = \int \left(\frac{1}{4\pi \mathscr{r}}\right) \big[\mu_0 \mathbf{J}(\mathbf{r}')\big]\, d\tau'. \tag{30}$$

This yields the final result:

$$\boxed{\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{\mathscr{r}} d\tau'.} \tag{31}$$

2.2 Multipole Expansion & Dipole Contribution

Now, we can use the same method just like what we did in the multipole expansion of the electric potential, and we substitute (13) into (31), we obtain the vector potential in a fixed current loop:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\oint \frac{\mathbf{I}}{\boldsymbol{\mathscr{r}}} d\ell' = \frac{\mu_0 I}{4\pi}\oint \frac{1}{\boldsymbol{\mathscr{r}}} d\boldsymbol{\ell}' = \frac{\mu_0 I}{4\pi}\sum_{n=0}^{\infty}\frac{1}{r^{n+1}}\oint (r')^n P_n(\cos\alpha)\, d\boldsymbol{\ell}'$$

$$= \frac{\mu_0 I}{4\pi}\left[\frac{1}{r}\oint d\boldsymbol{\ell}' + \frac{1}{r^2}\oint r'\cos\alpha\, d\boldsymbol{\ell}' + \cdots\right],$$

where $\oint d\boldsymbol{\ell}' = 0$, which means that there is no magnetic monopole, and $r'\cos\alpha = \mathbf{r}'\cdot\hat{\mathbf{r}}$. Likewise, the second term of the closed integral is called **magnetic dipole**, which can be written as

$$\mathbf{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0 I}{4\pi}\frac{1}{r^2}\oint \mathbf{r}'\cdot\hat{\mathbf{r}}\, d\boldsymbol{\ell}'. \tag{32}$$

By Green's Identities and Some Integrals (iii) (let $\mathbf{c} = \hat{\mathbf{r}}$), the integral in (32) equals:

$$\oint \mathbf{r}'\cdot\hat{\mathbf{r}}\, d\boldsymbol{\ell} = \left(\int d\mathbf{a}'\right)\times\hat{\mathbf{r}}. \tag{33}$$

Define the magnetic dipole moment $\mathbf{m}$ to be

$$\mathbf{m} := I\int d\mathbf{a}'. \tag{34}$$

Combining (32) to (34), we can finally obtain the dipole vector potential in the magnetic field:

$$\mathbf{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi}\frac{\mathbf{m}\times\hat{\mathbf{r}}}{r^2} \tag{35}$$

In general, as the same reason of the arbitrary orientation of the sources, the dipole vector potential can also be written as:

$$\mathbf{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi}\frac{\mathbf{m}\times\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}. \tag{36}$$

The dipole moment of the infinitesimal volume is $\mathbf{m} = \mathbf{M}d\tau$, where $\mathbf{M}$ is the magnetic dipole moment per unit volume. So, the dipole vector potential is now

$$\boxed{\mathbf{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi}\int \frac{\mathbf{M}(\mathbf{r}')\times\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2} d\tau'.} \tag{37}$$

So, what does these dipole potentials mean? Look up "*Polarization & Magnetization, Yen-Ting Wu.*" We will continue to discuss the properties of dipoles there.

Some Identities

Triple Products:

$$\begin{cases} \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) \\ \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \quad \text{Bac(k)-Cab law} \end{cases}$$

Products Rules:

$$\begin{cases} \boldsymbol{\nabla}(fg) = f(\boldsymbol{\nabla} g) + g(\boldsymbol{\nabla} f) \\ \boldsymbol{\nabla}(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\boldsymbol{\nabla} \times \mathbf{B}) + \mathbf{B} \times (\boldsymbol{\nabla} \times \mathbf{A}) + (\mathbf{A} \cdot \boldsymbol{\nabla})\mathbf{B} + (\mathbf{B} \cdot \boldsymbol{\nabla})\mathbf{A} \\ \boldsymbol{\nabla} \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\boldsymbol{\nabla} \times \mathbf{A}) - \mathbf{A} \cdot (\boldsymbol{\nabla} \times \mathbf{B}) \\ \boldsymbol{\nabla} \cdot (f\mathbf{A}) = f(\boldsymbol{\nabla} \cdot \mathbf{A}) + \mathbf{A} \cdot \boldsymbol{\nabla} f \\ \boldsymbol{\nabla} \times (f\mathbf{A}) = f(\boldsymbol{\nabla} \times \mathbf{A}) - \mathbf{A} \times \boldsymbol{\nabla} f \\ \boldsymbol{\nabla} \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \boldsymbol{\nabla})\mathbf{A} - (\mathbf{A} \cdot \boldsymbol{\nabla})\mathbf{B} + \mathbf{A}(\boldsymbol{\nabla} \cdot \mathbf{B}) - \mathbf{B}(\boldsymbol{\nabla} \cdot \mathbf{A}) \end{cases}$$

Second Derivatives:

$$\begin{cases} \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \mathbf{A}) = 0 \\ \boldsymbol{\nabla} \times (\boldsymbol{\nabla} f) = \mathbf{0} \\ \boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \mathbf{A}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \end{cases}$$

Green's Identities and Some Integrals:

$$
\begin{cases}
\displaystyle\int_{\mathcal{V}} \left(T\,\nabla^2 U - U\nabla^2 T\right) d\tau = \oint_{\mathcal{S}} \left(T\boldsymbol{\nabla} U - U\boldsymbol{\nabla} T\right)\cdot d\mathbf{a} \\
\displaystyle\int_{\mathcal{S}} \boldsymbol{\nabla} T \times d\mathbf{a} = -\oint_{\mathcal{P}} T\,d\boldsymbol{\ell} \\
\displaystyle\oint (\mathbf{c}\cdot\mathbf{r})\,d\boldsymbol{\ell} = \left(\int d\mathbf{a}\right)\times\mathbf{c} \\
\displaystyle\int_{\mathcal{V}} \boldsymbol{\nabla}\times\mathbf{F}\,d\tau = -\oint_{\mathcal{S}} \mathbf{F}\times d\mathbf{a}
\end{cases}
$$

Delta functions:

$$
\begin{cases}
\boldsymbol{\nabla}\cdot\left(\dfrac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}\right) = 4\pi\delta^3(\boldsymbol{\mathscr{r}}) \\
\boldsymbol{\nabla}\left(\dfrac{1}{\mathscr{r}}\right) = \boldsymbol{\nabla}\left(\dfrac{1}{\mathbf{r}-\mathbf{r}'}\right) = -\dfrac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2} \\
\boldsymbol{\nabla}'\left(\dfrac{1}{\mathscr{r}}\right) = \boldsymbol{\nabla}'\left(\dfrac{1}{\mathbf{r}-\mathbf{r}'}\right) = \dfrac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2} \\
\nabla^2\left(\dfrac{1}{\mathscr{r}}\right) = -4\pi\delta^3(\boldsymbol{\mathscr{r}})
\end{cases}
$$